

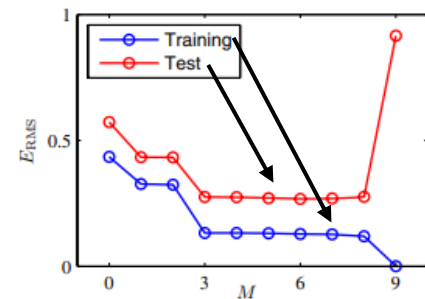
BITS PILANI DUBAI CAMPUS
INTERNATIONAL ACADEMIC CITY, DUBAI
ACADEMIC-UNDERGRADUATE STUDIES DIVISION
BITS F464- MACHINE LEARNING – TEST 1
FIRST SEMESTER 2020-2021

TIME: 50 MINUTES
DATE: 24.10.2020, Sa8

MARKS : 20

QUESTIONS	MARKS
1	1
2	1
3	2
4	3
5	3
6	5
7	5

1. Given below is the RMS chart for a model evaluated over training and test set. Comment over the model

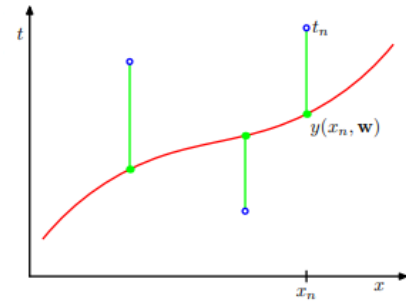


2. In childhood we used play the game (pictured beside) with some tricks. Identify the learning algorithm that well suits to play this game. In a line or two explain the learning algorithm



3. Given the figure with 'x' representing the independent data and 't' representing the target data. The line y denotes the predicted line and the dots symbolize the target data points.

- What formulae you arrive at when you square the distance covered by the vertical lines?
- Pen down the formulae and mention where it is used.
- Extend the formula by adding a penalty term.



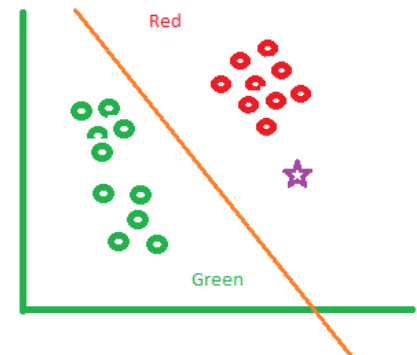
- Differentiate one-versus-one and one-versus rest discriminant function.
 - Explain with diagram.
 - Can they be applied for multi class classification?
 - For 592 classes, find out how many classifiers are needed for one-versus-one and one-versus rest discriminant function.
- Assume you build a machine learning model for prediction. Of course, you will construct a gradient descent graph to assess the model.
 - What is a gradient descent graph. What would be its axis legends.
 - How do you adjust the learning rate?
 - In what factor the variation in learning rate affects the gradient.
 - How to reach the minimal error using gradient.

Explain with a neat diagram.

6. Assume there are 10 green balls and 9 red balls. The balls are perfectly classified by a linear decision boundary as shown in figure. Given a star point, find to which class the point belongs to.

Explain in detail with derivations and equations.

NOTE: Do not use probability or measure of central tendency or measure of deviation



- Think of a scenario that projects the mean of two group of data points onto W vector, such that the two group of data points are as widely separated as possible. Derive a scale dependent formula on which W vector depends and a scale independent formula as well.
Explain with a diagram.
Write the steps in detail.

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BITS F464- MACHINE LEARNING – TEST 2
FIRST SEMESTER 2020-2021

TIME: 50 MINUTES

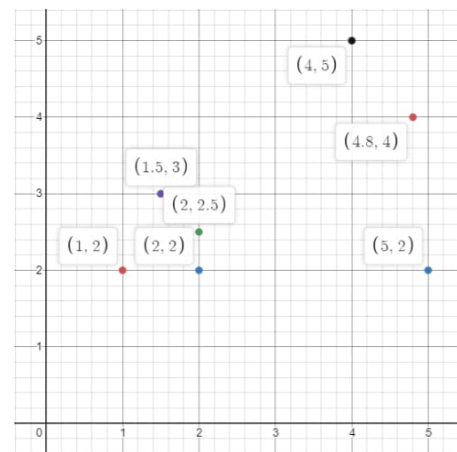
MARKS : 20

DATE: 25.11.2020, W8

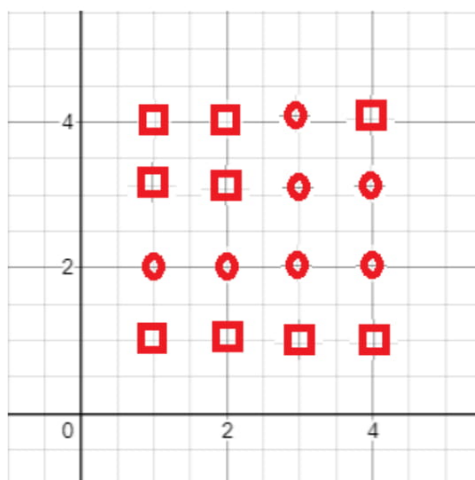
Each question carries 5 marks.

1. Points of two different classes are put in a 2D space. The task is to separate the points of two classes as wider as possible. Find the width of the separation plane that you would like to construct and also the W vector.

Class 1	Class 2
(1,2)	(4,5)
(2,2)	(4.8,4)
(2,2.5)	(5,2)
(1.5,3)	



2. Given below is the graph with two set of points (oval and square). Draw a complete decision tree separating the two class of points. Apply cost complexity pruning and prune the branches of the tree. Calculate the alpha values every iteration. Finally write the decision conditions and alpha value. Run only two iterations. Consider the horizontal axis as x axis and vertical axis as y axis in graph.



3. Analyze the datasets given below and the attributes provided. Calculate the entropy of each of them. Which of the datasets have high entropy and low entropy? Why do you calculate entropy and where it is used?

DATASET 1:

ID	ATT1	ATT2	ATT2	CLASS LABEL
1	Y	l	120	n
2	N	m	100	n
3	Y	s	130	n
4	N	l	450	n
5	Y	m	120	n
6	n	s	120	n
7	y	l	130	n
8	n	m	500	n
9	y	s	600	n
10	n	l	700	y

DATASET 2:

ID	ATT1	ATT2	ATT2	CLASS LABEL
1	y	l	120	n
2	n	m	100	n
3	y	s	130	n
4	n	l	450	n
5	y	m	120	n
6	n	s	120	n
7	y	l	130	n
8	n	m	500	n
9	y	s	600	n
10	n	l	700	y

DATASET 3:

ID	ATT1	ATT2	ATT2	CLASS LABEL
1	y	l	120	y
2	n	m	100	y
3	y	s	130	y
4	n	l	450	y
5	y	m	120	y
6	n	s	120	y
7	y	l	130	n
8	n	m	500	n
9	y	s	600	n
10	n	l	700	n

DATASET 4:

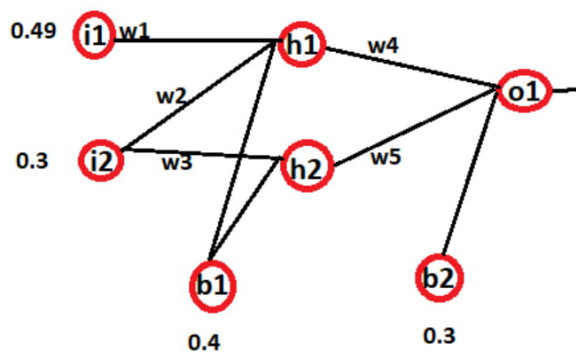
ID	ATT1	ATT2	ATT2	CLASS LABEL
1	y	l	120	n
2	n	m	100	n
3	y	s	130	n
4	n	l	450	n
5	y	m	120	n
6	n	s	120	y
7	y	l	130	y
8	n	m	500	y
9	y	s	600	y
10	n	l	700	y

DATASET 5:

ID	ATT1	ATT2	ATT2	CLASS LABEL
1	y	l	120	n
2	n	m	100	n
3	y	s	130	n
4	n	l	450	N
5	y	m	120	N
6	n	s	120	n
7	y	l	130	n
8	n	m	500	n
9	y	s	600	n
10	n	l	700	n

4. Given a neural network. Calculate the error at the output node. Apply Back propagation algorithm with learning rate as 0.9 and calculate the new weight of w5 after one iteration.

w1=0.4, w2=0.3, w3=0.2, w4=0.1, w5=0.2. Target value at o1=0.89



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ACADEMIC-UNDERGRADUATE STUDIES DIVISION
BITS F464- MACHINE LEARNING
COMPREHENSIVE EXAMINATION- SESSION 1
FIRST SEMESTER 2020-2021

TIME: 80 MINUTES (08:00 AM TO 09:20 AM)
DATE: 30.12.2020 FN

MARKS: 20

QUESTION	MARKS
1	2
2	2
3	5
4	5
5	6

QUESTION 1

What are the factors/parameters to be considered to split two different class of data points using decision boundary line? Assume you are working with projected values of data points. **(2 marks)**

QUESTION 2

Explain least confident and smallest margin uncertainty sampling with formula. **(2 marks)**

QUESTION 3

Given: The states. The maximal reward of reaching the Goal State is 5000.

- Write the formulae to calculate the optimal discounted cumulative reward value. **(0.5 mark)**
- Find the cumulative discount reward of all the states from A to I if the discounted reward factor is 0.8. **(2 mark)**
- Write the formulae to choose the optimal action using Q-learning. **(0.5 mark)**
- Find the value of each action taken to move from one state to another. **(2 mark)**

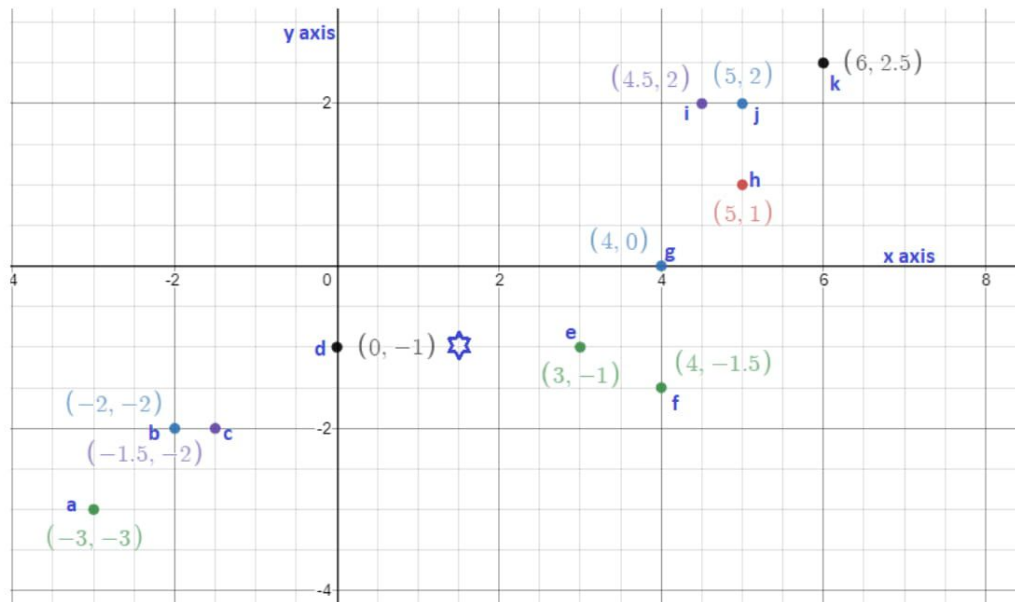
A	B	GOAL STATE
C	D	E
F	G	H
		I

QUESTION 4

Given the points, labels of the points and the axis.

- Apply 2-NN classification with centroid values as $(-1.6, -2)$ and $(4.5, 0.7)$ and classify the points to the respective clusters. Find to which category the “star point” belongs to, by identifying the nearest neighbor algorithm. **(2 marks)**
- Apply 3-NN classification with centroid values as $(-1.6, -2)$, $(3.66, -0.83)$, $(5.12, 1.875)$ and classify the points to the respective clusters. Find to which category the “star point” belongs to, by identifying the nearest neighbor algorithm. **(3 marks)**

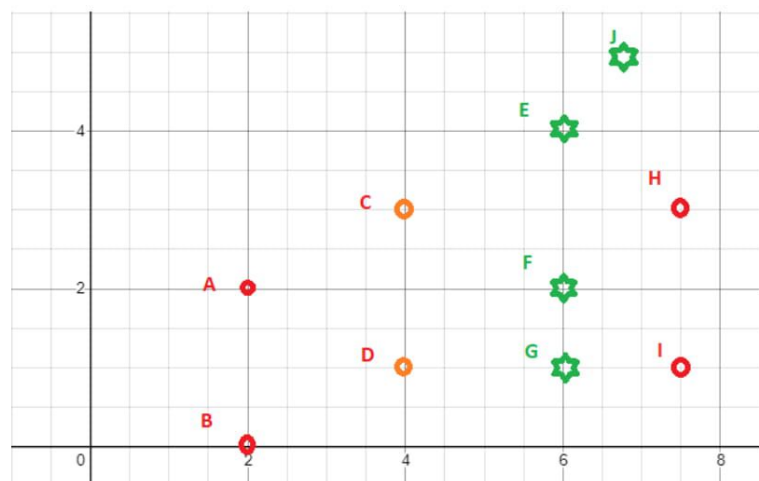
Use only K-NN formulae. Don't use brute force approach.



QUESTION 5

Given the graph. Identify a machine learning approach and build a suitable strong classifier by strategically choosing optimal number of feasible classifiers. Also identify the power of each feasible classifiers and derive the final hypothesis equation for the strong classifier. Limit the number of classifiers to two. **(6 marks)**

NOTE: Step by step calculation is a must. Use the x and y axis values to label the points in the graph for your better understanding.



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BITS F464- MACHINE LEARNING
COMPREHENSIVE EXAMINATION- SESSION 2
FIRST SEMESTER 2020-2021

TIME: 90 MINUTES (09:30 AM TO 11:00 AM)
DATE: 30.12.2020 FN

MARKS: 20

QUESTION	MARKS
6	2
7	2
8	5
9	5
10	6

QUESTION 6

Mention the penalty equation for linear and logistic regression. **(1 mark)**

Can you apply linear regression's penalty equation for logistic function? If so what sort of gradient model would be generated. Draw the gradient curve and Justify the same. **(1 mark)**

QUESTION 7

Given table

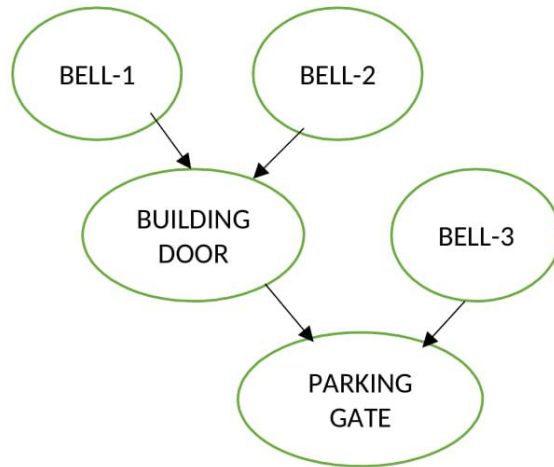
Attribute 1	Attribute 2	Attribute 3	Class Label
.	.	.	.
.	.	.	.
.	.	.	.

Say you are asked to frame three regression equations

1. One-dimensional regression equations
2. Two-dimensional regression equations
3. Three-dimensional regression equations

Write the derived regression equation for all the three models mentioned above. **(2 mark)**

QUESTION 8



Scenario: The building door works based on the rings of bell-1 and bell-2. The parking gate works based on the working of building door and rings of bell 3.

1. Bell-1 rings with probability 0.9 and Bell-2 rings with probability 0.8 and Bell-3 rings with probability 0.5.

2. Working of building door in terms of Bell-1 and Bell-2 behavior is given here in table below:

Bell-1	Bell-2	P(Building door opening)
0	0	0.7
0	1	0.4
1	0	0.3
1	1	0.1

3. Working of parking gate in terms of building door and Bell-3 behavior is given here in table below:

Building door (0 refers to close and 1 refers to open)	Bell-3	P(Parking Gate opening)
0	0	0.8
0	1	0.2
1	0	0.2
1	1	0.1

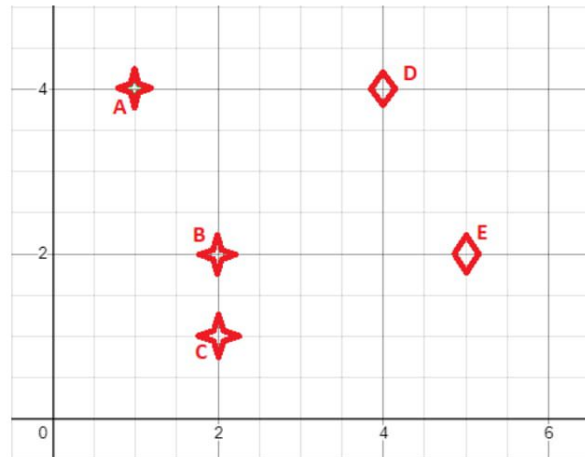
- Write the joint distribution function of the model. **(1 mark)**
- Find the probability of parking gate being open and close **(2 marks)**
- Find the conditional probability that the building door is closed given parking gate is open i.e. $P(\text{building door is closed} / \text{parking gate is open})$ **(2 marks)**

NOTE: Write in steps the formulae and calculations done.

QUESTION 9

Write the formulae to calculate the width of the SVM hyperplane. Identify the support vectors from the below mentioned graph and calculate the W vector. Calculate the width of the SVM. **(5 marks)**

NOTE: Use the x and y axis values to label the points in the graph. Step by step calculation is a must.



QUESTION 10

Given the data set apply a supervised dimensionality reduction approach and find the new values of all the data points using discriminant equation. Is there any change in points allocated to its respective class label? **(6 marks)**

x	y	CLASS
1	1	CLUSTER 1
0.5	1	CLUSTER 1
1	0.5	CLUSTER 1
-1	-1	CLUSTER 2
-0.5	-1	CLUSTER 2
-1	-0.5	CLUSTER 2